

SHATHURSIMA RAVEENDRAN

Second Year University Undergraduate Student

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Profile

I am a dedicated computer engineering enthusiast with a deep curiosity about the world of networks and connectivity. Driven by a passion for applying knowledge to explore and innovate, I am committed to continually learning and deepening my understanding of the computer world. Whether it's solving problems, creating solutions, or understanding new technologies, I approach every endeavour with enthusiasm, a growth mindset, and a desire to contribute meaningfully to the evolving landscape of technology.

Education

University of Peradeniya	2024 – 2028
Second year Computer Engineering Undergraduate GPA : 3.5/4.0	
J/Vembadi Girls' High School	2014 – 2023
Secondary Education	

Experience

Casual Instructor	2025 – Present
Department of Computer Engineering, University of Peradeniya	
Design Team Member (2026/2027) & Editorial Team Member (2025/2026)	
WIE Affinity Group of Student Branch, UOP	
Editorial Team Member	2026 – 2027
SEDS PERA	
OGV IR Member	2025
AIESEC in Kandy	

Projects

Dementia Risk Prediction Model

- Developed a machine learning-based project during a hackathon to predict the likelihood of dementia using non-medical factors (demographic, lifestyle, and cognitive data).
- Implemented a custom Logistic Regression model from scratch using NumPy, optimizing parameters via gradient descent.
- Managed end-to-end data cleaning, normalization, model serialization with Pickle, and created an interactive prediction module.
- *Skills:* Machine Learning, Python, NumPy, Teamwork

ARM Project — Mini Command-Line Shell (CO1020)

- Developed a functional command-line shell in ARM32 Assembly supporting terminal commands like `hello`, `help`, `clear`, `exit`, `len`, and `upper`.
- Implemented low-level input/output handling, memory management, and system calls to simulate a minimal Linux shell environment.
- *Skills:* ARM Assembly, Computer Systems Programming

Libtiny3d — 3D Graphics Engine from Scratch

- Built a lightweight 3D software rendering engine in C from the ground up without high-level graphics APIs (OpenGL/DirectX).
- Programmed a complete rendering pipeline featuring a custom math library for 3D transformations, DDA line drawing, viewport clipping, and Lambertian lighting.
- *Skills:* C, Makefile, Linear Algebra, 3D Geometry

Signal Intelligence Dashboard — SignalMon

- Built an automated open-source intelligence system to monitor, process, and visualize real-time socio-economic and operational signals from Sri Lankan news sources.

- Extracted actionable strategic insights from unstructured text data using NLP frameworks and robust engineering pipelines.
- *Skills:* Natural Language Processing, Data Engineering

Volunteering

Coordinator for "CODE WITH WIE2025" · IEEE Women in Engineering

Invigilator & Technical Team Member · ICPC Foundation Contest 2025

Educational Volunteer · WIE Affinity Group of Student Branch, UOP

Volunteer · Yarl IT Hub

Skills, Certifications & Languages

Skills: Strategic Planning, Problem Solving, Team Management, ML/AI, Software Development, Basic Networking, Content Writing, Canva, Microsoft Office.

Languages: Tamil (Native), English (Fluent), Sinhala (Basic).

Certifications:

- Get Started with Databricks for Machine Learning (Simplilearn)
- Google AI Essentials (Google)
- Java Programming (Great Learning)
- Introduction to Python (Sololearn)
- Python for Beginners (University of Moratuwa)